

Consider the problem of solving two 8-puzzles.

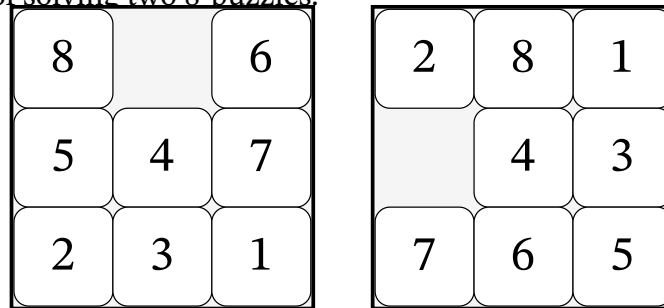


Figure 1: Sample 8-puzzles

1. Determine the size of the reachable state space.

3
(CO1)
(PO1)

Solution:

Each puzzle has $9!/2$ reachable states. The joint state space has $(9!)^2/4$ reachable states

Rubric:

- 1 point for single puzzle
- 1 point for both
- 1 point for considering half the states are reachable

2. Formulate the scenario as a game by identifying the start state, action, successor function, and goal test.

4
(CO2)
(PO2)

Solution:

Start State: Two arbitrary 8-puzzle states

Action: Move one empty piece of one puzzle horizontally or vertically

Successor function: one move on an unsolved puzzle

Goal test: Both puzzles are solved

Rubric:

- 1 point for each

3. Suppose we make the problem adversarial by considering two players who take turns moving. Before each move, a coin is flipped to determine the puzzle on which to make a move in that turn. The winner is the first to solve one puzzle.

2 + 6
(CO3)
(PO3)

- a) Recommend an algorithm for finding the optimal move in this game.

Solution:

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Rubric:

- 2 points for choosing the correct one

- b) “The game will never end if both players play perfectly.” - Do you agree with the statement? Justify your position.

Solution:

I agree.

Consider a state in which the coin has come up heads, say, and you get to work on a puzzle that is 2 steps from the goal. Should you move one step closer? If you do, your opponent wins if he tosses heads; or if he tosses tails, you toss tails, and he tosses heads; or any sequence where both toss tails n times and then he tosses heads. So his probability of winning is at least $1/2 + 1/8 + 1/32 + \dots = 2/3$. So it seems you are better off moving away from the goal.

Rubric:

- 2 points for correct decision
- 4 points for justification